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11.2.3 Reduction to bidiagonal form

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place zeros right here. So what if we instead we compute a different Householder transform --let's call it H_0 hat -- in such a way that this matrix that result it has zeros right here. So what we do is we look at this vector right here. Well it's a row vector but we could pretend that it's a column vector because when we apply from the right to a row, that's the same as applying the transpose from the left to

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Homework 11.2.3.1

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Let $B \in \mathbb{R}^{m \times m}$ be a bidiagonal matrix:

$$B = \begin{pmatrix} \beta_{0,0} & \beta_{0,1} & 0 & \cdots & 0 & 0 \\ 0 & \beta_{1,1} & \beta_{1,2} & \cdots & 0 & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \beta_{m-2,m-2} & \beta_{m-2,m-1} \\ 0 & 0 & 0 & \cdots & 0 & \beta_{m-1,m-1} \end{pmatrix}.$$

Show that $T = B^T B$ is a tridiagonal symmetric matrix.

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Solution:

For a complete solution see the [notes](#).

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